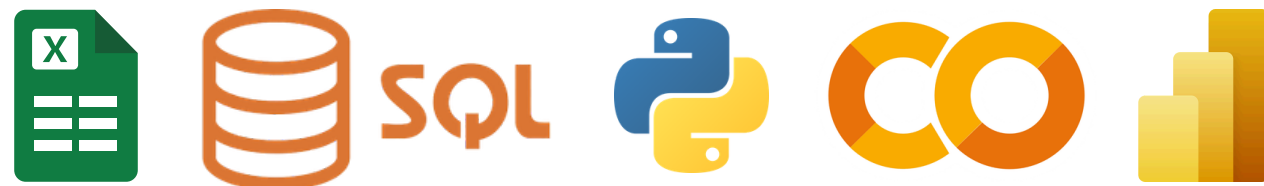




Unlocking Sales Performance Through Data Insights

Utari W Ardhana

Role: Data Analyst



January - 2026



Table of Content



01

Overview

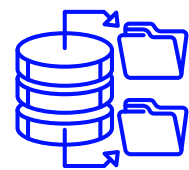
Penjelasan secara general terkait dengan tema yang dipilih beserta pengerjaan



02

5W + 1H

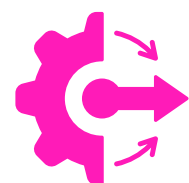
Pertanyaan dan jawaban terkait dengan data set dan tema



03

Sumber data & Flow

Penjelasan sumber data dan source sumber data yang dipertanggung jawabkan & Rancangan alur pengerjaan



04

Proses & Output

Penjelasan proses eksekusi proyek dan Hasil dari eksekusi data



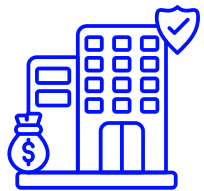
05

Summary

Rangkuman dari hasil yang telahdikerjakan

Overview

Business Understanding



Perusahaan memiliki data penjualan, namun belum dimanfaatkan secara optimal

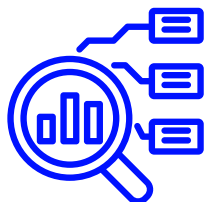


Pengambilan keputusan masih berbasis intuisi



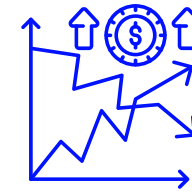
Tidak ada visibilitas performa penjualan secara real-time

Objectives



Mengolah data penjualan menjadi insight yang actionable

Key Problems



Penjualan produk yang tidak stabil tiap periode



Tidak diketahui produk mana yang paling profitable



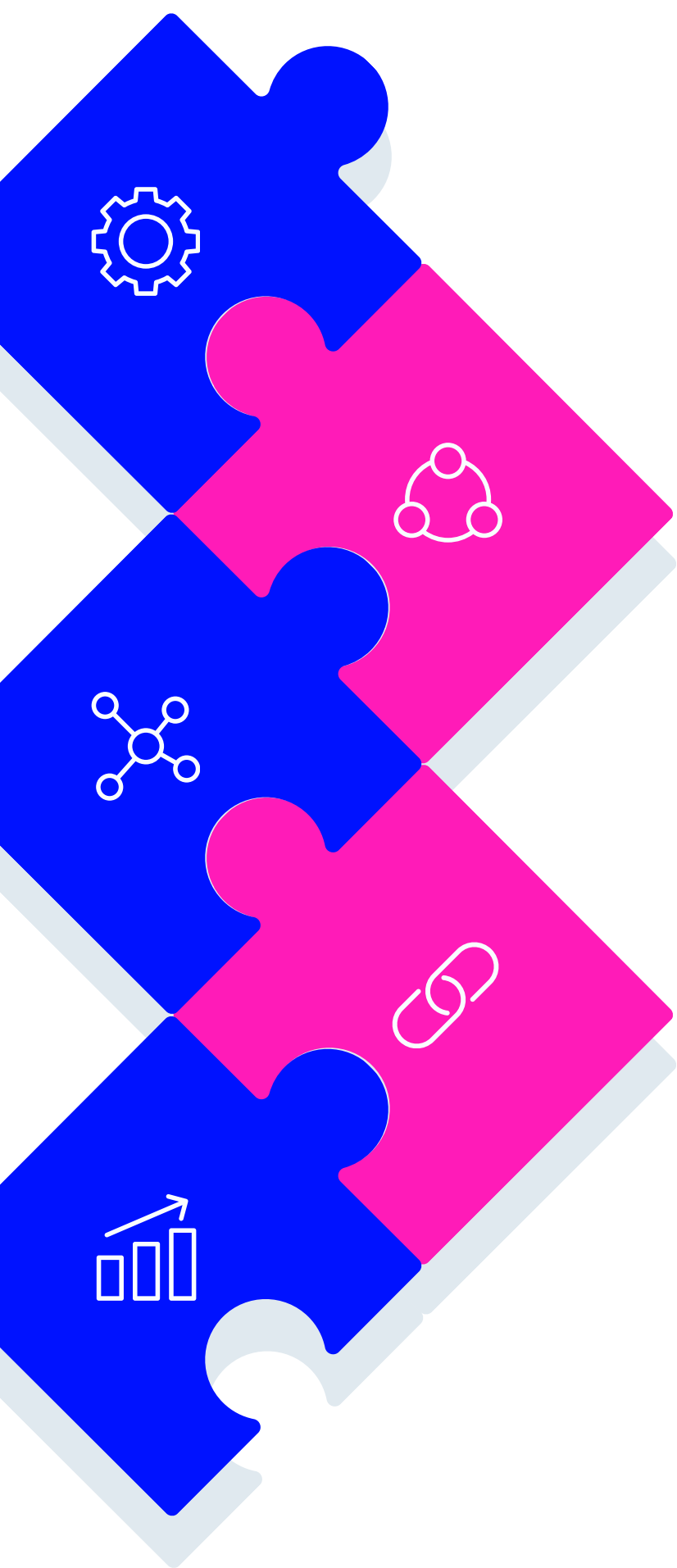
Performa tiap region/customer tidak terukur



Tidak ada insight untuk meningkatkan revenue



Membangun dashboard simple untuk melihat performance sales



5W + 1H

What (Apa yang dianalisis?)

Proyek ini berfokus pada analisis performa penjualan berdasarkan data historis untuk mengidentifikasi pola, tren, serta faktor-faktor yang mempengaruhi kinerja bisnis.

Why (Mengapa analisis ini penting?)

Banyak organisasi memiliki data penjualan yang besar, namun belum dimanfaatkan secara optimal untuk mendukung pengambilan keputusan.

Who (Siapa yang akan menggunakan hasilnya?)

Dashboard dan insight yang dihasilkan ditujukan untuk berbagai stakeholder, antara lain, Tim Sales Marketing untuk memonitor performa penjualan dan target & data analyst

When (Kapan data dianalisis?)

Data yang digunakan merupakan data historis dalam periode 3 Tahun

Where (Dari mana data berasal?)

Data berasal dari sumber dataset penjualan, baik berupa Dataset publik (Kaggle)

How (Bagaimana proses analisis dilakukan?)

Data Collection, Data Cleaning, Data Transformation, Data Analysis, Data Visualization, Dashboard Developmentu

Data Source

 MURSALEEN AMEER · UPDATED 2 YEARS AGO

▲ 79

<> Code

Download

CRM  + Sales  + Opportunities 



CRM(Customer Relationship Management) Opportunities

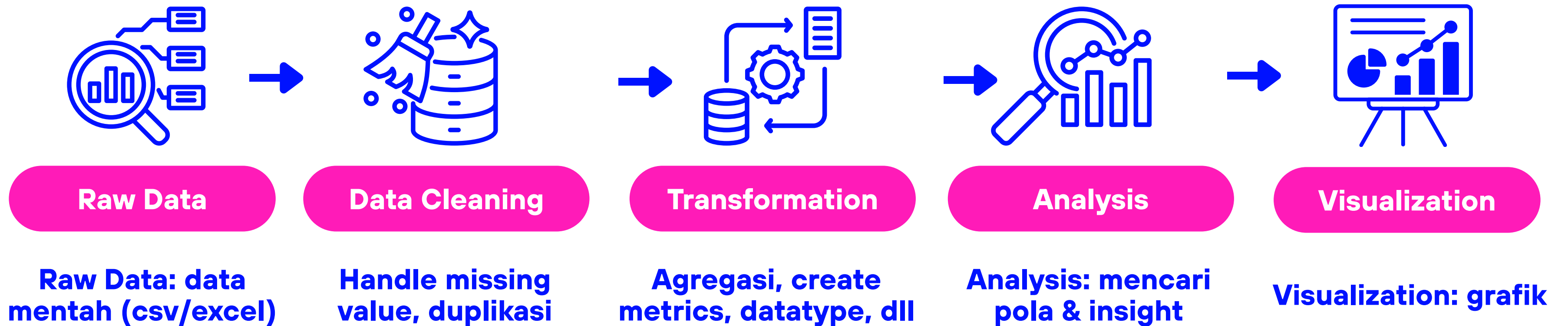
kaggle

Sumber data:

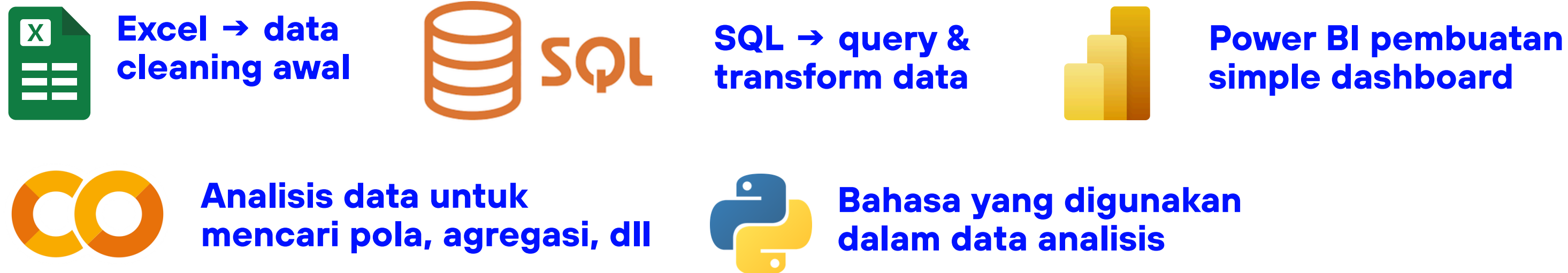
<https://www.kaggle.com/datasets/innocentmfa/crm-sales-opportunities>

 Views	 Downloads	 Engagement	 Comments
22.8K  1109 in the last 30 days	5852  279 in the last 30 days	0.25612 downloads per view	0 posted

Data Flow



Tools & Tech Stack



Proses Raw Data

	A	B	C	D	E	F	G	H	I	J
1	opportunity_id	sales_agent	product	account	deal_stage	engage_date	close_date	close_value		
2	1C1I7A6R	Moses Frase	GTX Plus Basic	Cancity	Won	2016-10-20	2017-03-01	1054		
3	Z063OYW0	Darcel Schlecht	GTXPro	Isdom	Won	2016-10-25	2017-03-11	4514		
4	EC4QE1BX	Darcel Schlecht	MG Special	Cancity	Won	2016-10-25	2017-03-07	50		
5	MV1LWRNH	Moses Frase	GTX Basic	Codehow	Won	2016-10-25	2017-03-09	588		
6	PE84CX4O	Zane Levy	GTX Basic	Hatfan	Won	2016-10-25	2017-03-02	517		
7	ZNBS69V1	Anna Snelling	MG Special	Ron-tech	Won	2016-10-29	2017-03-01	49		
8	9ME3374G	Vicki Laflamme	MG Special	J-Texon	Won	2016-10-30	2017-03-02	57		
9	7GN8Q4LL	Markita Hansen	GTX Basic	Cheers	Won	2016-11-01	2017-03-07	601		
10	OLK9LKZB	Niesha Huffines	GTX Plus Basic	Zumgoity	Won	2016-11-01	2017-03-03	1026		
11	HAXMC4IX	James Ascencio	MG Advanced		Engaging	2016-11-03				
12	NL3JZH1Z	Anna Snelling	MG Special	Bioholding	Won	2016-11-04	2017-03-10	53		
13	KWVA7VR1	Gladys Colclough	GTXPro	Genco Pura Olive Oil Company	Lost	2016-11-04	2017-03-18	0		
14	S8DX3XOU	James Ascencio	GTX Plus Pro	Sunnamplex	Won	2016-11-04	2017-03-10	5169		
15	ENB2XD8G	Maureen Marcano	GTX Plus Pro	Sonron	Won	2016-11-04	2017-03-06	4631		
16	09YE9QOV	Hayden				2017-03-11	3393			

accounts.csv

	A	B	C	D	E	F
1	sales_agent	manager	regional_office			
2	Anna Snelling	Dustin Brinkmann	Central			
3	Cecily Lampkin	Dustin Brinkmann	Central			
4	Versie Hillebrand	Dustin Brinkmann	Central			
5	Lajuana Vencill	Dustin Brinkmann	Central			
6	Moses Frase	Dustin Brinkmann	Central			
7	Jonathan Berthelot	Melvin Marxen	Central			
8	Marty Freudenburg	Melvin Marxen	Central			
9	Gladys Colclough	Melvin Marxen	Central			
10	Niesha Huffines	Melvin Marxen	Central			
11	Darcel Schlecht	Melvin Marxen	Central			
12	Mei-Mei Johns	Melvin Marxen	Central			
13	Violet Mclelland	Cara Losch	East			
14	Corliss Cosme	Cara Losch	East			
15	Rosie Papadopoulos	Cara Losch	East			
16	Garret Kinder	Cara Losch	East			
17	Wilburn Farren	Cara Losch	East			
18	Elizabeth Anderson	Cara Losch	East			

data_dictionary.csv

	A	B	C
1	product	series	sales_price
2	GTX Basic	GTX	550
3	GTX Pro	GTX	4821
4	MG Special	MG	55
5	MG Advanced	MG	3393
6	GTX Plus Pro	GTX	5482
7	GTX Plus Basic	GTX	1096
8	GTX 500	GTX	36768

products.csv

	A	B	C	D	E	F
1	Table	Field	Description			
2	accounts	account	Company name			
3	accounts	sector	Industry			
4	accounts	year_established	Year Established			
5	accounts	revenue	Annual revenue (in millions of USD)			
6	accounts	employees	Number of employees			
7	accounts	office_location	Headquarters			
8	accounts	subsidiary_of	Parent company			
9	products	product	Product name			
10	products	series	Product series			
11	products	sales_price	Suggested retail price			
12	sales_teams	sales_agent	Sales agent			
13	sales					
14	sale					

sales_pipeline.csv

	A	B	C	D	E	F	G	H
1	account	sector	year_established	revenue	employees	office_location	subsidiary_of	
2	Acme Corporation	technology	1996	1100.04	2822	United States		
3	Betasoloin	medical	1999	251.41	495	United States		
4	Betatech	medical	1986	647.18	1185	Kenya		
5	Bioholding	medical	2012	587.34	1356	Philipines		
6	Bioplex	medical	1991	326.82	1016	United States		
7	Blackzim	retail	2009	497.11	1588	United States		
8	Bluth Company	technology	1993	1242.32	3027	United States	Acme Corporation	
9	Bubba Gump	software	2002	987.39	2253	United States		
10	Cancity	retail	2001	718.62	2448	United States		
11	Cheers	entertainment	1993	4269.9	6472	United States	Massive Dynamic	
12	Codehow	software					Corporation	
13	Condax	medical						

sales_teams.csv

Proses Data Cleaning

Import library untuk digunakan

```
[1] 1 import pandas as pd
    2 import matplotlib.pyplot as plt
    3 import numpy as np
    4 import seaborn as sns
    5 from scipy import stats
```

```
1 df_products.isnull().sum()
```

```
0
product 0
series  0
sales_price 0
```

dtype: int64

Melakukan pengecekan data null dari semua dataframe

```
1 df_sales['account'] = df_sales['account'].fillna('Unknown')
2 df_sales['engage_date'] = df_sales['engage_date'].fillna('Unknown')
3 df_sales['close_date'] = df_sales['close_date'].fillna('Unknown')
4 df_sales['close_value'] = df_sales['close_value'].fillna(0)
5
6 print("Null values after filling:")
7 print(df_sales.isnull().sum())
```

```
... Null values after filling:
opportunity_id    0
sales_agent       0
product           0
account           0
deal_stage        0
engage_date       0
close_date        0
close_value       0
dtype: int64
```

Memastikan duplikasi data

```
1 df_accounts.duplicated().sum()

np.int64(0)
```

```
1 df_accounts['subsidiary_of'] = df_accounts['subsidiary_of'].fillna(0)
2 df_accounts.head()
3 df_accounts['subsidiary_of'].isnull().sum()
```


Membuat Master Data dengan Merge df_accounts, df_dict, df_products, df_sales, df_tim

▼ Merge Sales and Accounts

[]

1 df_master = pd.merge(df_sales, df_accounts, on='account', how='inner')

2 df_master.head()

▼

	opportunity_id	sales_agent	product	account	deal_stage	engage_date	close_date	close_value	sector	year_established	revenue	employees	office_location
0	1C1I7A6R	Moses Frase	GTX Plus Basic	Cancity	Won	2016-10-20	2017-03-01	1054.0	retail	2001	718.62	2448	United States
1	Z063OYW0	Darcel Schlecht	GTXPro	Isdom	Won	2016-10-25	2017-03-11	4514.0	medical	2002	3178.24	4540	United States
2	EC4QE1BX	Darcel Schlecht	MG Special	Cancity	Won	2016-10-25	2017-03-07	50.0	retail	2001	718.62	2448	United States
3	MV1LWRNH	Moses Frase	GTX Basic	Codehow	Won	2016-10-25	2017-03-09	588.0	software	1998	2714.90	2641	United States
4	PE84CX4O	Zane Levy	GTX Basic	Hatfan	Won	2016-10-25	2017-03-02	517.0	services	1982	792.46	1299	United States

1 df_master = pd.merge(df_master, df_products, on='product', how='inner')

2 df_master.head()

	opportunity_id	sales_agent	product	account	deal_stage	engage_date	close_date	close_value	sector	year_established	revenue	employees	office_location	series	sales_price
0	1C1I7A6R	Moses Frase	GTX Plus Basic	Cancity	Won	2016-10-20	2017-03-01	1054.0	retail	2001	718.62	2448	United States	GTX	1096
1	EC4QE1BX	Darcel Schlecht	MG Special	Cancity	Won	2016-10-25	2017-03-07	50.0	retail	2001	718.62	2448	United States	MG	55
2	MV1LWRNH	Moses Frase	GTX Basic	Codehow	Won	2016-10-25	2017-03-09	588.0	software	1998	2714.90	2641	United States	GTX	550
3	PE84CX4O	Zane Levy	GTX Basic	Hatfan	Won	2016-10-25	2017-03-02	517.0	services	1982	792.46	1299	United States	GTX	550
4	ZNBS69V1	Anna Snelling	MG Special	Ron-tech	Won	2016-10-29	2017-03-01	49.0	medical	1992	3922.42	6837	United States	MG	55

Hasil merge dan terbuat Master Data

```
1 df_master.describe()
```

	close_value	year_established	revenue	employees	sa
count	6117.000000	6117.000000	6117.000000	6117.000000	6
mean	1061.787804	1995.582802	2452.836668	5639.306523	1
std	2104.520914	9.195814	2579.883527	6725.775735	2
min	0.000000	1979.000000	4.540000	9.000000	
25%	0.000000	1988.000000	646.100000	1238.000000	
50%	56.000000	1995.000000	1618.890000	3194.000000	1
75%	1062.000000	2002.000000	2952.730000	7523.000000	3
max	30288.000000	2017.000000	11698.030000	34288.000000	26

```
1 df_master.info()
```

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 6117 entries, 0 to 6116
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   opportunity_id         6117 non-null   object 
1   sales_agent            6117 non-null   object 
2   product                6117 non-null   object 
3   account                6117 non-null   object 
4   deal_stage            6117 non-null   object 
5   engage_date            6117 non-null   object 
6   close_date             6117 non-null   object 
7   close_value            6117 non-null   float64
8   sector                 6117 non-null   object 
9   year_established       6117 non-null   int64  
10  revenue                6117 non-null   float64
11  employees               6117 non-null   int64  
12  office_location        6117 non-null   object 
13  series                 6117 non-null   object 
14  sales_price            6117 non-null   int64  
15  manager                6117 non-null   object 
16  regional_office        6117 non-null   object 
dtypes: float64(2), int64(3), object(12)
memory usage: 812.5+ KB
```

Exploratory Analysis

1. Melihat opportunity, won dan convention rate dari df_master dan kolom deal_stage

```
Exploratory Analysis

1 total_opp = df_master.shape[0]
2 won_opp = df_master[df_master['deal_stage'] == 'Won'].shape[0]
3
4 conversion_rate = (won_opp / total_opp) * 100
5
6 print(f"Total opp: {total_opp}")
7 print(f"Won opp: {won_opp}")
8 print(f"Conversion Rate: {conversion_rate:.2f}%")

Total opp: 6117
Won opp: 3509
Conversion Rate: 57.36%
```

2. Ingin mengetahui Avg Sales berdasarkan engage_date dan close_date

```
1 df_master['engage_date'] = pd.to_datetime(df_master['engage_date'], errors='coerce')
2 df_master['close_date'] = pd.to_datetime(df_master['close_date'], errors='coerce')
3
4 sales_cycle = df_master['close_date'] - df_master['engage_date']
5 average_sales_cycle = sales_cycle.mean()
6
7 print(f"Average Sales Cycle: {average_sales_cycle}")

Average Sales Cycle: 48 days 10:47:00.992092020
```

3. Mengubah format date agar sesuai dengan format PowerBI

```
1 # Replace 'NaT' strings (from previous datetime conversion and formatting) with 'Unknown'
2 df_master['engage_date'] = df_master['engage_date'].replace('NaT', 'Unknown')
3 df_master['close_date'] = df_master['close_date'].replace('NaT', 'Unknown')
4
5 # Fill any remaining NaN values with 'Unknown'
6 df_master['engage_date'] = df_master['engage_date'].fillna('Unknown')
7 df_master['close_date'] = df_master['close_date'].fillna('Unknown')
8
9 print("Date columns after cleaning 'NaT' values and filling NaNs:")
10 print(df_master[['engage_date', 'close_date']].head())
11 print("\nNull counts after cleaning:")
12 print(df_master[['engage_date', 'close_date']].isnull().sum())

... Date columns after cleaning 'NaT' values and filling NaNs:
   engage_date  close_date
0  20-10-2016  01-03-2017
1  25-10-2016  07-03-2017
2  25-10-2016  09-03-2017
3  25-10-2016  02-03-2017
4  29-10-2016  01-03-2017

Null counts after cleaning:
engage_date    0
close_date     0
dtype: int64
```

4. Mengetahui Revenue terbesar dari setiap kantor regional

```
1 revenue_by_tim = df_master.groupby(['regional_office', 'manager'])['revenue'].sum().sort_values(ascending=False)
2 display(revenue_by_tim)
```

		revenue
regional_office	manager	
West	Summer Sewald	3348224.04
Central	Dustin Brinkmann	2650522.83
West	Celia Rouche	2638097.18

5. Create KPI: Total Sales, Total Orders, Average Order Value.

```
1 print("### KPI Analysis ###\n")
2
3 # Filter for 'Won' deals as sales typically refer to successful opportunities
4 df_won_deals = df_master[df_master['deal_stage'] == 'Won'].copy()
5
6 # Convert close_date to datetime for monthly analysis, coercing errors for 'Unknown' values
7 df_won_deals['close_date_dt'] = pd.to_datetime(df_won_deals['close_date'], format='%d-%m-%Y', errors='coerce')
8
9 # Drop rows where close_date is 'Unknown' or invalid after conversion for this analysis
10 df_won_deals_valid_dates = df_won_deals.dropna(subset=['close_date_dt']).copy()
11
12 # --- KPI per Month ---
13 print("--- KPIs per Month ---")
14 df_won_deals_valid_dates['close_month'] = df_won_deals_valid_dates['close_date_dt'].dt.to_period('M')
15 kpi_monthly = df_won_deals_valid_dates.groupby('close_month').agg(
16     total_sales=('close_value', 'sum'),
17     total_orders=('opportunity_id', 'count')
18 ).reset_index()
19 kpi_monthly['average_order_value'] = kpi_monthly['total_sales'] / kpi_monthly['total_orders']
20 print(kpi_monthly.sort_values('close_month'))
21 print("\n")
22
23 # --- KPI per Product Category (Series) ---
24 print("--- KPIs per Product Category (Series) ---")
25 kpi_category = df_won_deals.groupby('series').agg(
26     total_sales=('close_value', 'sum'),
27     total_orders=('opportunity_id', 'count')
28 ).reset_index()
29 kpi_category['average_order_value'] = kpi_category['total_sales'] / kpi_category['total_orders']
30 print(kpi_category.sort_values('total_sales', ascending=False))
31 print("\n")
32
33 # --- KPI per Region ---
34 print("--- KPIs per Region ---")
35 kpi_region = df_won_deals.groupby('regional_office').agg(
36     total_sales=('close_value', 'sum'),
37     total_orders=('opportunity_id', 'count')
38 ).reset_index()
39 kpi_region['average_order_value'] = kpi_region['total_sales'] / kpi_region['total_orders']
40 print(kpi_region.sort_values('total_sales', ascending=False))
41 print("\n")
```

```
### KPI Analysis ###

--- KPIs per Month ---
   close_month  total_sales  total_orders  average_order_value
0    2017-03      757706.0           451      1680.057650
1    2017-04      449481.0           229      1962.799127
2    2017-05      673137.0           365      1844.210959
3    2017-06      884971.0           437      2025.105263
4    2017-07      443197.0           256      1731.238281
5    2017-08      696390.0           373      1866.997319
6    2017-09      746069.0           402      1855.893035
7    2017-10      466965.0           223      2094.013453
8    2017-11      622472.0           340      1830.800000
9    2017-12      754568.0           433      1742.651270

--- KPIs per Product Category (Series) ---
   series  total_sales  total_orders  average_order_value
1    GTX    3834189.0           2047      1873.077186
2    MG     2260155.0           1447      1561.959226
0    GTK     400612.0             15      26707.466667

--- KPIs per Region ---
   regional_office  total_sales  total_orders  average_order_value
2             West    2447825.0           1204      2033.077243
0           Central    2242184.0           1400      1601.560000
1             East    1804947.0            905      1994.416575
```


Metrics di PowerBI

```
1 Avg Deal Size =  
2 DIVIDE([Total Revenue], [Won Deals])
```

```
1 Lost Deals =  
2 CALCULATE(  
3     COUNTROWS(df_master),  
4     df_master[deal_stage] = "Lost"  
5 )
```

```
1 Total Opportunities = COUNT(df_master[opportunity_id])
```

```
1 Avg Sales Cycle = AVERAGE(df_master[sales_price])
```

```
1 Total Revenue = SUM(df_master[revenue])
```

```
1 Win Rate = DIVIDE([Won Deals], [Total Opportunities])
```

✓  df_master

☐ account

☐  Avg Deal Size

☐  Avg Sales Cycle

> ☐  close_date

☐ Σ close_value

☐  CustomDate

☐ deal_stage

☐ Σ employees

> ☐  engage_date

☐  Lost Deals

☐ manager

☐ office_location

☐ opportunity_id

☐ product

☐ regional_office

☐ Σ revenue

☐ sales_agent

☐ Σ sales_price

☐ sector

☐ series

☐  Total Opportunities

☐  Win Rate

☐  Won Deals

Dashboard Overview

DASHBOARD SALES PERFORMANCE



Key Insights (Data-Driven Findings)

- **Revenue Performance**

1. Total revenue mencapai 1.33M dengan total 3,509 deals, menunjukkan volume transaksi yang tinggi
2. Namun, win rate 57.36% mengindikasikan masih ada ruang untuk optimasi konversi

- **Monthly Sales Trend**

1. Terdapat fluktuasi penjualan tiap bulan
2. Beberapa bulan menunjukkan peak performance, mengindikasikan adanya faktor musiman atau campaign tertentu
3. **Insight: perlu identifikasi strategi yang digunakan pada bulan dengan performa tertinggi untuk direplikasi**

- **Product Performance**

1. Produk dengan kontribusi tertinggi mendominasi revenue secara signifikan dibanding produk lain
2. Terdapat gap yang cukup besar antara produk top performer dan low performer
3. **Insight:**
 - a. Fokus scaling produk unggulan
 - b. Evaluasi strategi untuk produk dengan performa rendah

- **Sales Team Performance**

1. Performa antar sales agent cukup bervariasi
2. Beberapa agent memiliki:
 - a. **win rate tinggi + deal besar → high performer**
 - b. **win rate rendah atau cycle lama → perlu improvement**
 - c. **Insight: best practice dari top performer bisa dijadikan benchmark tim**

- **Product Performance**

1. Rata-rata sales cycle cukup panjang ($\pm 1.89Rb$)
2. Terdapat variasi signifikan antar agent
3. Insight:
 - a. **Optimasi proses closing**
 - b. **Identifikasi bottleneck di pipeline**

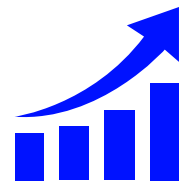
- **Deal Stage Bottleneck**

1. Distribusi deal menunjukkan mayoritas masih berada di tahap tertentu (misalnya negotiation/prospecting)
2. Insight:
 - a. **Perlu strategi untuk mendorong deal ke tahap closing**
 - b. **Evaluasi funnel conversion rate**

- **Correlation Insight (Scatter Plot)**

1. Tidak semua agent dengan deal banyak memiliki win rate tinggi
2. Ada indikasi:
 - a. **Trade-off antara volume dan kualitas deal**
 - b. **Insight: penting untuk balancing antara quantity vs quality**

Turning Insight into Action



Increase Revenue Opportunity

Dengan meningkatkan win rate sebesar 5–10%, perusahaan berpotensi meningkatkan revenue secara signifikan tanpa menambah jumlah leads



Faster Sales Cycle = Better Cashflow

Optimasi sales cycle dapat mempercepat proses closing dan meningkatkan efisiensi operasional



Better Resource Allocation

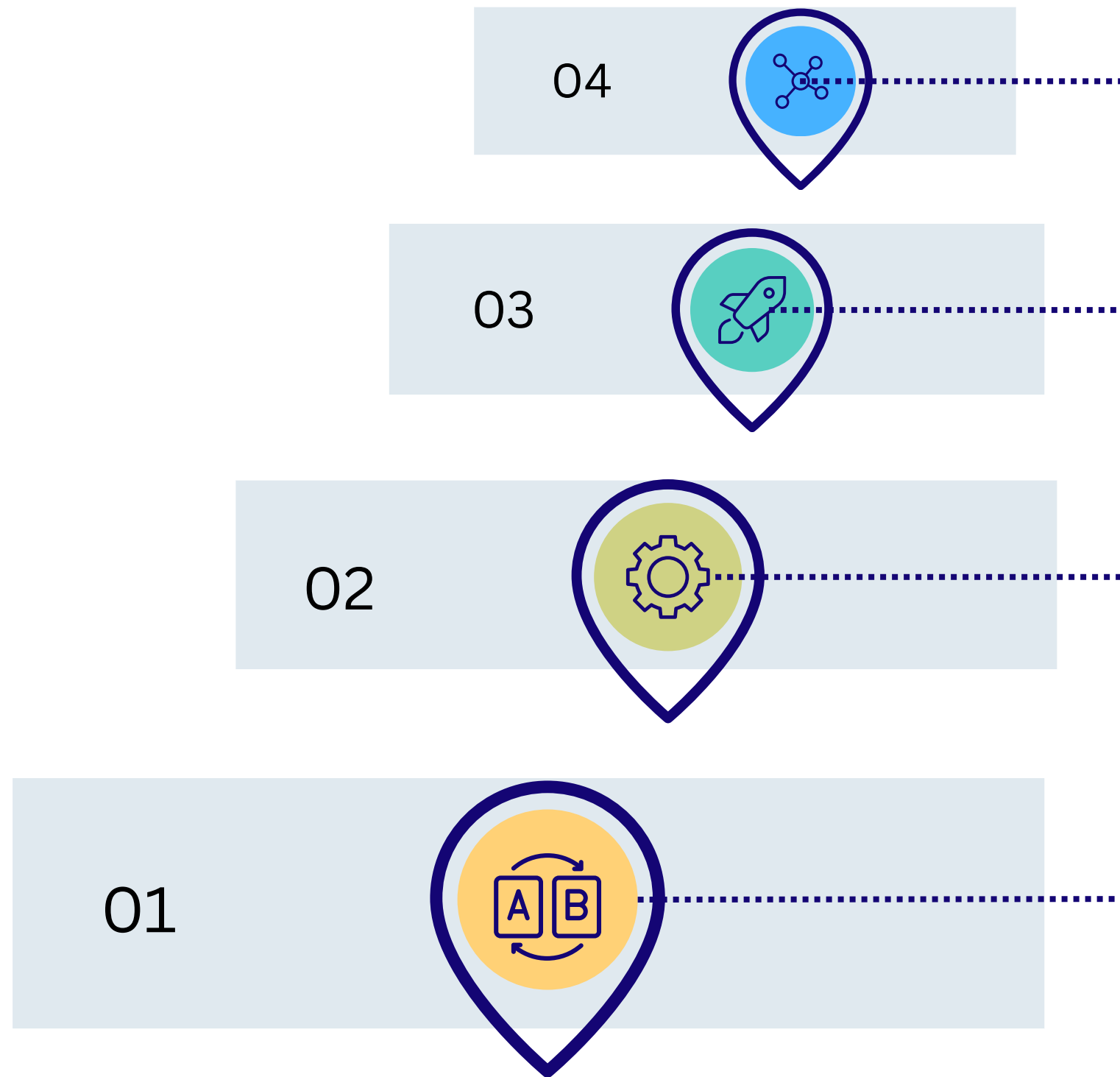
Fokus pada produk dan region dengan performa tinggi untuk memaksimalkan ROI



Sales Team Optimization

Replikasi strategi top performer untuk meningkatkan performa tim secara keseluruhan

From Dashboard to Decision



Deep Dive Analysis

- Analisis lebih lanjut per produk, region, dan customer segment
- Identifikasi root cause dari low performance

Strategy Implementation

- Optimasi funnel di tahap yang menjadi bottleneck
- Replikasi strategi dari high-performing sales

Performance Monitoring

- Gunakan dashboard sebagai tools monitoring harian/mingguan
- Track KPI secara berkala

Data Enhancement

- Tambahkan data tambahan (customer behavior, marketing campaign, dll.)
- Integrasi dengan sistem real-time



Thank You !

Let's Collaborate!

 @uutwdh

 utariwardhanaa@gmail.com

 bhaddbaeby

